

Research Notes on Protein structure prediction

Protein structure prediction

>> Section 1

Background Early methods of secondary structure prediction, introduced in the 1960s and early 1970s, focused on identifying likely alpha helices and were based mainly on helix-coil transition models. Significantly more accurate predictions that included beta sheets were introduced in the 1970s and relied on statistical assessments based on probability parameters derived from known solved structures. These methods, applied to a single sequence, are typically at most about 60–65% accurate, and often underpredict beta sheets. Since the 1980s, artificial neural networks have been applied to the prediction of protein structures. The evolutionary conservation of secondary structures can be exploited by simultaneously assessing many homologous sequences in a multiple sequence alignment, by calculating the net secondary structure propensity of an aligned column of amino acids. In concert with larger databases of known protein structures and modern machine learning methods such as neural nets and support vector machines, these methods can achieve up to 80% overall accuracy in globular proteins. The theoretical upper limit of accuracy is around 90%, partly due to idiosyncrasies in DSSP assignment near the ends of secondary structures, where local conformations vary under native conditions but may be forced to assume a single conformation in crystals due to packing constraints. Moreover, the typical secondary structure prediction methods do not account for the influence of tertiary structure on formation of secondary structure; for example, a sequence predicted as a likely helix may still be able to adopt a beta-strand conformation if it is located within a beta-sheet region of the protein and its side chains pack well with their neighbors. Dramatic conformational changes related to the protein's function or environment can also alter local secondary structure.

>> Section 2

Homology modeling is based on the reasonable assumption that two homologous proteins will share very similar structures. Because a protein's fold is more evolutionarily conserved than its amino acid sequence, a target sequence can be modeled with reasonable accuracy on a very distantly related template, provided that the relationship between target and template can be discerned through sequence alignment. It has been suggested that the primary bottleneck in comparative modelling arises from difficulties in alignment rather than from errors in structure prediction given a known-good alignment. Unsurprisingly, homology modelling is most accurate when the target and template have similar sequences. Protein threading scans the amino acid sequence of an unknown structure against a database of solved structures. In each

case, a scoring function is used to assess the compatibility of the sequence to the structure, thus yielding possible three-dimensional models. This type of method is also known as 3D-1D fold recognition due to its compatibility analysis between three-dimensional structures and linear protein sequences. This method has also given rise to methods performing an inverse folding search by evaluating the compatibility of a given structure with a large database of sequences, thus predicting which sequences have the potential to produce a given fold.

>> Section 3

Evolutionary covariation to predict 3D contacts As sequencing became more commonplace in the 1990s several groups used protein sequence alignments to predict correlated mutations and it was hoped that these coevolved residues could be used to predict tertiary structure (using the analogy to distance constraints from experimental procedures such as NMR). The assumption is when single residue mutations are slightly deleterious, compensatory mutations may occur to restabilize residue-residue interactions. This early work used what are known as local methods to calculate correlated mutations from protein sequences, but suffered from indirect false correlations which result from treating each pair of residues as independent of all other pairs. In 2011, a different, and this time global statistical approach, demonstrated that predicted coevolved residues were sufficient to predict the 3D fold of a protein, providing there are enough sequences available (>1,000 homologous sequences are needed). The method, EVfold, uses no homology modeling, threading or 3D structure fragments and can be run on a standard personal computer even for proteins with hundreds of residues. The accuracy of the contacts predicted using this and related approaches has now been demonstrated on many known structures and contact maps, including the prediction of experimentally unsolved transmembrane proteins.

>> Section 4

Protein structure and terminology Proteins are chains of amino acids joined together by peptide bonds. Many conformations of this chain are possible due to the rotation of the main chain about the two torsion angles ϕ and ψ at the $C\alpha$ atom. This conformational flexibility is responsible for differences in the three-dimensional structure of proteins.

>> Section 5

A great number of software tools for protein structure prediction exist. Approaches include homology modeling, protein threading, ab initio methods, secondary structure prediction, and transmembrane helix and signal peptide

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prediction. In particular, deep learning based on long short-term memory has been used for this purpose since 2007, when it was successfully applied to protein homology detection and to predict subcellular localization of proteins. Some recent successful methods based on the CASP experiments include I-TASSER, HHpred and AlphaFold. In 2021, AlphaFold was reported to perform best. Knowing the structure of a protein often allows functional prediction as well. For instance, collagen is folded into a long-extended fiber-like chain and it makes it a fibrous protein. Recently, several techniques have been developed to predict protein folding and thus protein structure, for example, Itasser, and AlphaFold.

>> Section 6

The peptide bonds in the chain are polar, i.e. they have separated positive and negative charges (partial charges) in the carbonyl group, which can act as hydrogen bond acceptor and in the NH group, which can act as hydrogen bond donor. These groups can therefore interact in the protein structure. Proteins consist mostly of 20 different types of L- α -amino acids (the proteinogenic amino acids). These can be classified according to the chemistry of the side chain, which also plays an important structural role. Glycine takes on a special position, as it has the smallest side chain, only one hydrogen atom, and therefore can increase the local flexibility in the protein structure. Cysteine in contrast can react with another cysteine residue to form one cystine and thereby form a cross link stabilizing the whole structure. Protein structure arises from a sequence of secondary structure elements, such as α helices and β sheets. In secondary structures, regular patterns of H-bonds are formed between the main chain NH and CO groups of spatially neighboring amino acids, and the amino acids have similar Φ and ψ angles.